CO1-AT-1
COURSE CODE:DSA0502
COURSE NAME:query processing for data science
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1)Agriculture Crop Yield Analysis

Files:

- Farmers.csv
- Crops.csv
- Crops.json
- Crops.xml

Merge farmer and crop datasets, calculate average crop yield, identify top-performing

farmers, summarize district-wise production, and export the processed dataset.

Aim

To merge farmer and crop datasets, calculate the average crop yield, identify the top-performing farmers, summarize district-wise crop production, and export the processed dataset into a CSV file using Python.

Farmers.csv

Farmer_ID	Farmer_Name	District	Land_Area (Acres)
F101	Arun	Salem	5
F102	Priya	Erode	3
F103	Kumar	Salem	4
F104	Meena	Coimbatore	6

Farmer_ID	Farmer_Name	District	Land_Area (Acres)
F105	Ravi	Erode	2

Crops.csv / Crops.json / Crops.xml

Farmer_ID	Crop	Yield (Tons)	Production (Kg)
F101	Rice	12	12000
F102	Wheat	8	8000
F103	Sugarcane	15	15000
F104	Rice	18	18000
F105	Maize	7	7000

```
import pandas as pd

# Read datasets

farmers = pd.read_csv("Farmers.csv")

crops_csv = pd.read_csv("Crops.csv")

# Read JSON file

crops_json = pd.read_json("Crops.json")

# Read XML file

crops_xml = pd.read_xml("Crops.xml")

# Merge Farmers and Crop datasets

merged_data = pd.merge(farmers, crops_csv, on="Farmer_ID")

print("\nMerged Dataset")

print(merged_data)
```

```
# Average Crop Yield
avg_yield = merged_data["Yield (Tons)"].mean()
print("\nAverage Crop Yield:")
print(round(avg_yield, 2), "Tons")
# Top Performing Farmer
top_farmer = merged_data.loc[
    merged_data["Yield (Tons)"].idxmax()
]
print("\nTop Performing Farmer")
print(top_farmer)
# District-wise Production Summary
district_summary = merged_data.groupby("District")[
    "Production (Kg)"
].sum()
print("\nDistrict-wise Production")
print(district_summary)
# Export Processed Dataset
merged_data.to_csv("Processed_Crop_Data.csv", index=False)
print("\nProcessed dataset exported successfully.")
```

Merged Dataset

Farmer_ID	**Farmer_Name**	**District**	**Land_Area (Acres)**	**Crop**	**Yield (Tons)**	**Production (Kg)**
F101	Arun	Salem	5	Rice	12	12000
F102	Priya	Erode	3	Wheat	8	8000

Farmer_ID	Farmer_Name	District	Land_Area (Acres)	Crop	Yield (Tons)	Production (Kg)
F103	Kumar	Salem	4	Sugarcane	15	15000
F104	Meena	Coimbatore	6	Rice	18	18000
F105	Ravi	Erode	2	Maize	7	7000

Average Crop Yield:

12.0 Tons

Top Performing Farmer

Farmer_ID	Farmer_Name	District	Crop	Yield (Tons)
F104	Meena	Coimbatore	Rice	18

District-wise Production

District	Total Production (Kg)
Coimbatore	18000
Erode	15000
Salem	27000

QUE :2 University Courses

Files:

- **Courses.csv**

• Faculty.csv

• Courses.json

• Courses.xml

Task:

Read datasets from CSV, JSON, and XML formats. Inspect the datasets, handle missing values,

remove duplicate records, validate data types, merge related CSV datasets using a common key,

perform aggregation and summary statistics, identify the category with the highest value, and

export the cleaned dataset to a new CSV file.

Aim

To read university course datasets from CSV, JSON, and XML formats, inspect the datasets, handle missing values, remove duplicate records, validate data types, merge related datasets using a common key, perform aggregation and summary statistics, identify the department with the highest enrollment, and export the cleaned dataset into a new CSV file.

Input

Dataset Files

- Courses.csv
- Faculty.csv
- Courses.json
- Courses.xml

Courses.csv

Course_ID	Course_Name	Department	Faculty_ID	Students_Enrolled
C101	Data Structures	CSE	F001	60
C102	Database Systems	CSE	F002	55
C103	Digital Electronics	ECE	F003	45
C104	Thermodynamics	Mechanical	F004	50
C105	Structural Analysis	Civil	F005	40

Faculty.csv

Faculty_ID	Faculty_Name	Experience (Years)
F001	Dr. Kumar	12
F002	Dr. Priya	10
F003	Dr. Arun	8
F004	Dr. Meena	15
F005	Dr. Ravi	9

Courses.json / Courses.xml

Course_ID	Course_Name	Department	Faculty_ID	Students_Enrolled
C101	Data Structures	CSE	F001	60

Course_ID	Course_Name	Department	Faculty_ID	Students_Enrolled
C102	Database Systems	CSE	F002	55
C103	Digital Electronics	ECE	F003	45
C104	Thermodynamics	Mechanical	F004	50
C105	Structural Analysis	Civil	F005	40

```
import pandas as pd
# Read datasets
courses = pd.read_csv("Courses.csv")
faculty = pd.read_csv("Faculty.csv")
# Read JSON and XML files
courses_json = pd.read_json("Courses.json")
courses_xml = pd.read_xml("Courses.xml")
# Inspect datasets
print("Courses Dataset")
print(courses.head())
print("\nFaculty Dataset")
print(faculty.head())
# Handle Missing Values
courses.fillna("Not Available", inplace=True)
faculty.fillna("Not Available", inplace=True)
```

```
# Remove Duplicate Records
courses.drop_duplicates(inplace=True)
faculty.drop_duplicates(inplace=True)
# Validate Data Types
print("\nCourses Data Types")
print(courses.dtypes)
print("\nFaculty Data Types")
print(faculty.dtypes)
# Merge Datasets
merged_data = pd.merge(courses, faculty, on="Faculty_ID")
print("\nMerged Dataset")
print(merged_data)
# Summary Statistics
print("\nSummary Statistics")
print(merged_data["Students_Enrolled"].describe())
 # Department-wise Enrollment
department_summary = merged_data.groupby("Department")[
   "Students_Enrolled"
].sum()
print("\nDepartment-wise Enrollment")
print(department_summary)
# Department with Highest Enrollment
highest_department = department_summary.idxmax()
print("\nDepartment with Highest Enrollment:")
```

```
print(highest_department)
# Export Cleaned Dataset
merged_data.to_csv("Cleaned_University_Courses.csv", index=False)
print("\nCleaned dataset exported successfully.")
```

Output

Merged Dataset

Course_ID	Course_Name	Department	Faculty_ID	Students_Enrolled	Faculty_Name	Experience (Years)
C101	Data Structures	CSE	F001	60	Dr. Kumar	12
C102	Database Systems	CSE	F002	55	Dr. Priya	10
C103	Digital Electronics	ECE	F003	45	Dr. Arun	8
C104	Thermodynamics	Mechanical	F004	50	Dr. Meena	15
C105	Structural Analysis	Civil	F005	40	Dr. Ravi	9

Summary Statistics

count	5.000000
mean	50.000000
std	7.905694
min	40.000000

25%	45.000000
50%	50.000000
75%	55.000000
max	60.000000

Department-wise Enrollment

Department	Total Students Enrolled
CSE	115
Civil	40
ECE	45
Mechanical	50

Department with Highest Enrollment

CSE